

SEROV, Russian FederationWMO: **280440**Lat: **59.600N**Lon: **60.533E**Elev: **130**StdP: **99.77**Time Zone: **5.00 (E05)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -32.7	(c) -29.6	(d) -35.7	(e) 0.1	(f) -32.4	(g) -32.5	(h) 0.2	(i) -29.5	(j) 8.0	(k) -18.5	(l) 7.0	(m) -15.2	(n) 2.1	(o) 0	(p) 0.605

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.0	(c) 29.2	(d) 19.4	(e) 27.3	(f) 18.6	(g) 25.4	(h) 17.6	(i) 20.6	(j) 27.1	(k) 19.6	(l) 25.5	(m) 18.7	(n) 23.9	(o) 2.1	(p) 140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.4	13.5	22.6	17.5	12.7	22.0	16.5	12.0	21.3	59.7	27.6	56.7	25.4	53.5	24.0	23.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.3	6.3	5.4	-35.5	31.8	3.4	1.6	-38.0	33.0	-39.9	33.9	-41.8	34.8	-44.3	36.0
			-35.3	21.6	3.4	0.8	-37.7	22.2	-39.7	22.7	-41.7	23.1	-44.1	23.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	1.6	-16.1	-13.6	-4.5	2.7	10.2	15.4	17.6	15.3	9.1	2.4	-6.8	-13.1		
	DBStd	13.06	7.25	7.90	5.24	4.92	5.20	4.78	4.16	4.15	3.94	5.05	7.23	8.23		
	HDD10.0	3747	810	661	449	222	64	10	2	4	62	241	504	717		
	HDD18.3	6184	1068	894	707	469	257	111	65	112	276	494	754	976		
	CDD10.0	693	0	0	0	3	71	173	237	168	37	4	0	0		
	CDD18.3	89	0	0	0	0	4	24	42	18	1	0	0	0		
	CDH23.3	897	0	0	0	0	62	240	431	161	3	0	0	0		
	CDH26.7	227	0	0	0	0	6	50	122	49	0	0	0	0		
	WSAvg	2.2	1.8	2.1	2.5	2.6	2.5	2.3	1.9	1.9	2.1	2.3	2.1	1.9		
	PrecAvg	503	22	16	21	24	47	75	81	72	56	31	29	20		
(15) Precipitation	PrecMax	660	41	31	46	56	90	149	227	186	115	63	63	43		
	PrecMin	386	3	4	1	5	15	12	17	24	9	9	2	5		
	PrecStd	69	10	8	14	13	20	37	40	40	32	16	15	11		
	PrecStd	69	10	8	14	13	20	37	40	40	32	16	15	11		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.9	3.4	9.9	18.6	27.4	30.0	31.5	31.0	23.6	17.1	7.7	2.9		
		MCWB	-1.1	0.3	3.9	10.7	15.6	19.0	20.6	20.5	15.8	11.1	5.3	0.8		
	2%	DB	-2.2	1.2	6.3	14.5	25.1	28.1	29.7	27.6	20.1	14.0	5.2	1.4		
		MCWB	-3.5	-0.7	2.0	8.1	14.8	18.7	19.8	19.9	13.9	9.7	3.7	-0.3		
	5%	DB	-4.3	-0.6	4.4	12.4	22.6	26.3	27.9	24.5	17.3	11.7	3.2	-0.5		
		MCWB	-5.2	-2.3	0.7	6.4	13.4	17.9	19.3	18.0	12.8	8.4	1.6	-1.9		
	10%	DB	-6.3	-2.7	2.6	10.2	20.0	24.1	25.7	22.3	15.2	9.7	1.6	-2.5		
		MCWB	-7.2	-4.3	-0.2	5.4	12.0	16.5	18.2	16.9	12.0	7.1	0.1	-3.7		
	0.4%	WB	-0.8	0.9	4.5	11.5	17.0	21.1	21.7	21.3	16.6	11.7	5.7	1.1		
		MCDB	0.8	2.4	8.4	17.6	24.1	27.9	28.6	29.0	21.9	15.1	7.1	2.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	-3.3	-0.5	2.5	8.8	15.5	19.7	20.8	20.0	14.7	10.1	3.8	-0.3		
		MCDB	-2.3	1.2	5.6	13.2	23.1	25.7	27.2	26.1	18.1	13.4	5.1	1.2		
	5%	WB	-5.2	-2.3	1.2	7.0	14.3	18.7	20.0	18.8	13.5	8.5	1.8	-1.9		
		MCDB	-4.3	-0.6	3.7	11.7	21.5	24.8	26.1	23.9	16.3	11.6	3.0	-0.6		
	10%	WB	-7.2	-4.2	0.1	5.6	12.7	17.6	19.2	17.6	12.3	7.1	0.2	-3.6		
		MCDB	-6.3	-2.7	2.4	9.6	18.8	23.0	24.5	21.3	14.8	9.6	1.2	-2.5		
	5% DB	MDBR	6.5	8.6	8.7	8.7	11.1	10.0	10.0	8.8	7.7	6.2	5.8	6.1		
		MCDBR	5.7	7.7	9.7	12.3	14.8	13.0	13.5	12.8	11.0	9.2	5.4	5.2		
		MCWBR	5.1	6.2	6.6	7.7	7.3	5.6	4.8	5.7	5.8	5.6	4.4	4.6		
		MCWBR	5.6	7.0	8.4	11.5	13.8	12.1	11.8	11.5	8.9	8.7	5.3	5.3		
(35) Mean Daily Temperature Range	5% WB	MCWBR	5.1	5.7	6.2	7.6	7.5	5.9	4.7	5.5	5.1	5.6	4.5	4.7		
		MCWBR	5.1	5.7	6.2	7.6	7.5	5.9	4.7	5.5	5.1	5.6	4.5	4.7		
	taub	taub	0.266	0.300	0.354	0.392	0.392	0.392	0.413	0.415	0.367	0.329	0.281	0.253		
		taud	2.029	2.057	1.999	2.102	2.302	2.360	2.302	2.267	2.367	2.334	2.131	1.972		
(40) Clear-Sky Solar Irradiance	Ebn at Noon	Ebn at Noon	491	679	749	798	835	841	810	772	754	661	475	349		
		Edh at Noon	49	83	121	131	116	111	116	111	85	64	44	35		
	RadAvg	RadAvg	0.45	1.32	2.60	3.99	5.00	5.28	5.35	3.92	2.32	1.17	0.49	0.24		
(44) All-Sky Solar Radiation	RadStd	RadStd	0.04	0.14	0.24	0.42	0.43	0.40	0.45	0.39	0.28	0.16	0.07	0.03		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page